

Fall 2025 ECEN 629 Homework Set 4

Due Nov. 8th, 10:59 online

Prob. 1. Consider the problem

$$\begin{aligned} &\text{minimize: } x_1^2 + x_2^2 \\ &\text{subject to: } (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 \\ &\quad 2x_1 + x_2 \geq 3 \\ &\quad x_1 \geq 0, x_2 \geq 0. \end{aligned}$$

- (a) Derive the KKT conditions. Find a solution for these conditions. Is this the optimal solution?
- (b) Does the strong duality hold?
- (c) Compare the solution to the one you obtained last time using geometry. Are they the same?
- (d) Consider the problem

$$\begin{aligned} &\text{minimize: } x_1^2 + x_2^2 \\ &\text{subject to: } (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 + \mu_1 \\ &\quad 2x_1 + x_2 \geq 3 + \mu_2 \\ &\quad x_1 \geq 0, x_2 \geq 0. \end{aligned}$$

The optimal value is a function of μ_1 and μ_2 , i.e., $p^*(\mu_1, \mu_2)$. Compute

$$\left. \frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_1} \right|_{(0,0)} \quad \left. \frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_2} \right|_{(0,0)}.$$

Prob. 2. Consider the following optimization problem:

$$\begin{aligned} &\text{minimize: } \sum_{i=1}^n \alpha_i \max \left\{ \ln \frac{\sigma_i^2}{x_i}, 0 \right\} \\ &\text{subject to: } \sum_{i=1}^n \beta_i x_i \leq D, \end{aligned}$$

where $x = (x_1, x_2, \dots, x_n) \succ 0$ are the decision variables. Assume $\sigma^2 \succ 0$, $\alpha \succeq 0$, and $\beta \succ 0$.

- (a) Argue that this problem is convex;
- (b) Write down the KKT conditions;
- (c) Provide an explicit solution for this problem (in terms of certain intermediate variable), by providing a solution to the KKT conditions;

- (d) Show that equivalently, the inequality constraint can be replaced by an equality constraint for any $0 < D \leq \sum_{i=1}^n \beta_i \sigma_i^2$;
- (e) Let $n = 3$, $(\sigma_1^2, \sigma_2^2, \sigma_3^2) = (1, 4, 25)$, $\alpha = (1, 1, 2)$, and $\beta = (2, 1, 1)$. Consider the optimal value p^* as a function of the parameter D . Find this function in a closed form.
- (f) Design an algorithm to compute the optimal solution for any given $\sigma^2 \succ 0$, $\alpha \succeq 0$, and $\beta \succ 0$, with a computation complexity at most $O(n)$.

Prob. 3. Strong duality holds for the problem

$$\begin{aligned} \text{minimize: } & -4x_1^2 + x_2^2 + x_3^2 + 2(x_1 + x_2 + x_3) \\ \text{subject to: } & x_1^2 - 2x_2^2 + x_3^2 \leq 1, \end{aligned}$$

although the problem is not convex (see Appendix B.1). Derive the KKT conditions. Find all solutions for the KKT conditions. Which one is the optimal value (assuming primal and dual optimal values both attained)? You may need to use a program (Matlab, Mathematica, or Python package) to plot and solve a non-linear equation; approximate solution is acceptable.

Prob. 4. Download the CVX package (with either the Python or Matlab interface), read the user manual, and then answer the following questions. The following constraints will cause an error in CVX as they will not be recognized as valid constraints in a convex optimization problem. Convert them to valid forms, and explain your answer. You may need to read part of the user guide of CVX.

- (a) `norm([x + 2*y, x - y]) == 0;`
- (b) `norm([max(x,1), max(y,2)]) <= 3*x + y;`
- (c) `x^3 + y^3 <= 1; x >= 0; y >= 0;`
- (d) `x*y >= 1; x >= 0; y >= 0.`

Prob. 5. The gap function $\phi_r(x)$ measures the gap between the largest r component and smallest r component of the vector $x \in \mathbb{R}^n$, where $1 \leq r < n$. Consider the optimization problem

$$\begin{aligned} \text{minimize: } & \|Ax - b\| \\ \text{subject to: } & \phi_r(x) \leq \alpha. \end{aligned}$$

- (a) Find an equivalent LP formulation of the problem when the norm is ℓ_1 .
- (b) With the same ℓ_1 norm, set

$$A = \begin{bmatrix} 1 & 2 & 3 & 6 & 9 \\ 2 & 3 & 7 & 2 & 4 \\ 4 & 2 & 2 & 5 & 1 \end{bmatrix}$$

and $b = [8, -20, 9]^T$. Use CVX to solve this problem with $\alpha = 0.1, 1, 10$, and 100 , respectively, for $r = 1$, $r = 2$, and $r = 3$, respectively.

- (c) Consider the same problem with the difference that the norm is ℓ_3 norm. With the same matrix A and vector b above, solve the problem with CVX for various α values, and plot the optimal value p^* as a function of α , for $r = 1, 2$, and 3 .

Prob. 6. Consider the following problem

$$\begin{aligned} & \text{minimize: } \sum_{i=1}^k \phi(|r|_{[i]}) \\ & \text{subject to: } r = Ax - b, \end{aligned}$$

where $|r|_{[1]} \geq |r|_{[2]} \geq \dots \geq |r|_{[m]}$ are the numbers $|r|_1, |r|_2, \dots, |r|_m$ sorted in decreasing order. Here matrix A is $m \times n$.

- (a) When $\phi(u) = u$, reformulate the problem into an LP. The problem with this penalty function minimizes the sum of the largest k residuals. (Hint: you need the formulation in exercise 5.19.)
- (b) When $\phi(u)$ is the deadzone-linear penalty function

$$\phi(u) = \begin{cases} 0 & |u| \leq a; \\ |u| - a & |u| > a, \end{cases}$$

where $a > 0$, reformulate the problem into an LP. This penalty function minimizes the sum of the largest k residuals, but with a deadzone. Recall we have discussed a problem similar to this in class.